

Hybrid Quantum - Classical Machine Learning for Autoencoder-Based Wireless Communication Systems

Motivation

Current Status

- Machine learning is becoming an important tool in wireless communication systems.
- End-to-end autoencoder models can jointly optimize the transmitter and receiver.

Problems

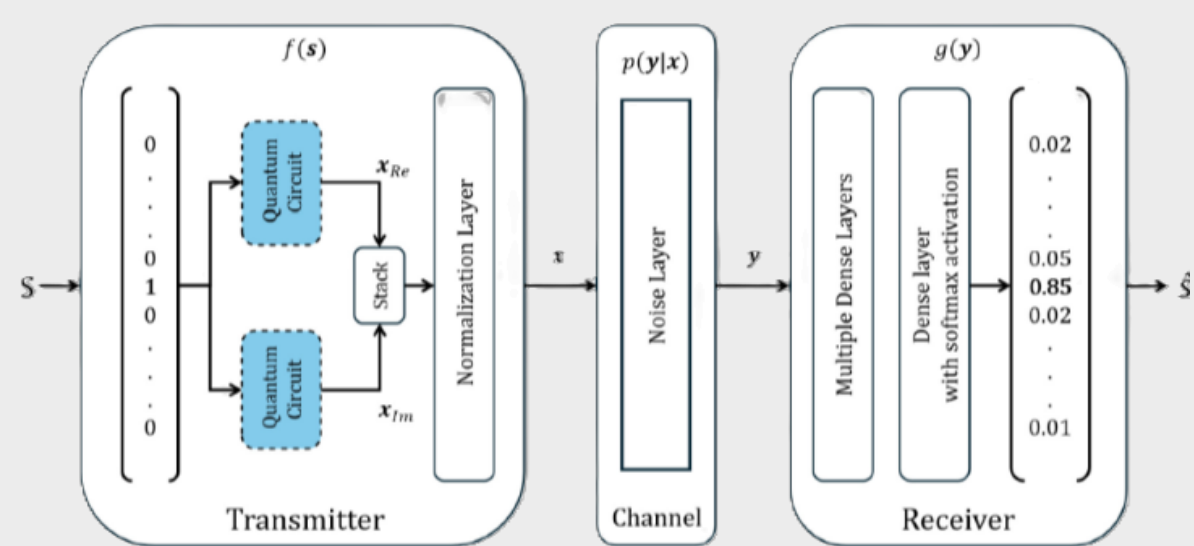
- As autoencoder-based systems scale, the number of trainable parameters increases.
- Higher computational complexity.
- Reduction of memory efficiency

Solution: Hybrid Autoencoders

Hybrid Autoencoder

Transmitter:
Quantum Encoder (QNN)

Receiver:
Classical Decoder (CNN)



Quantum Encoder

Embedding :

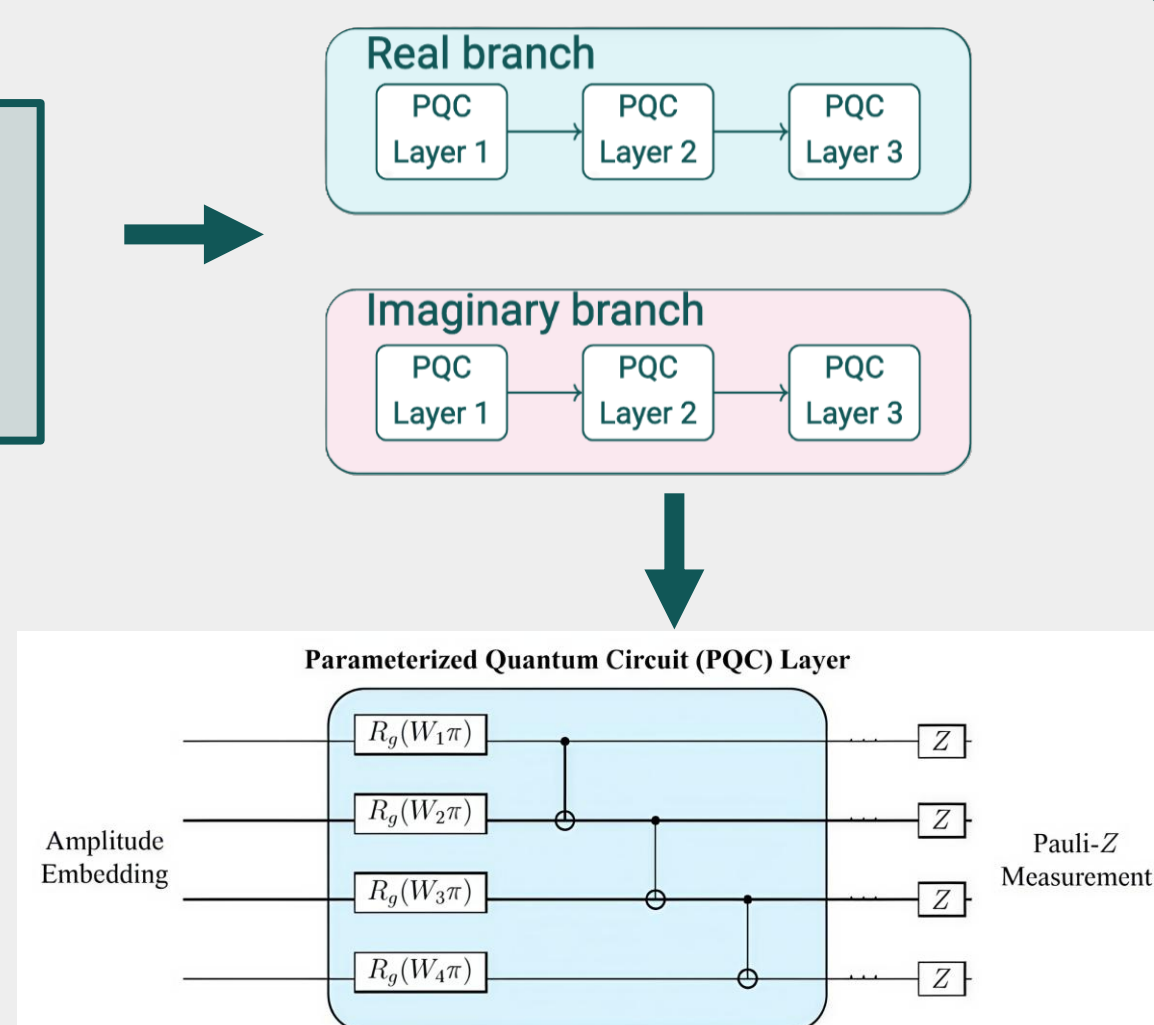
Converts information to quantum bits.

TVB: Ry rotations to all qubits

Entanglement: through CNOT gates.

Measurement: Pauli-Z basis.

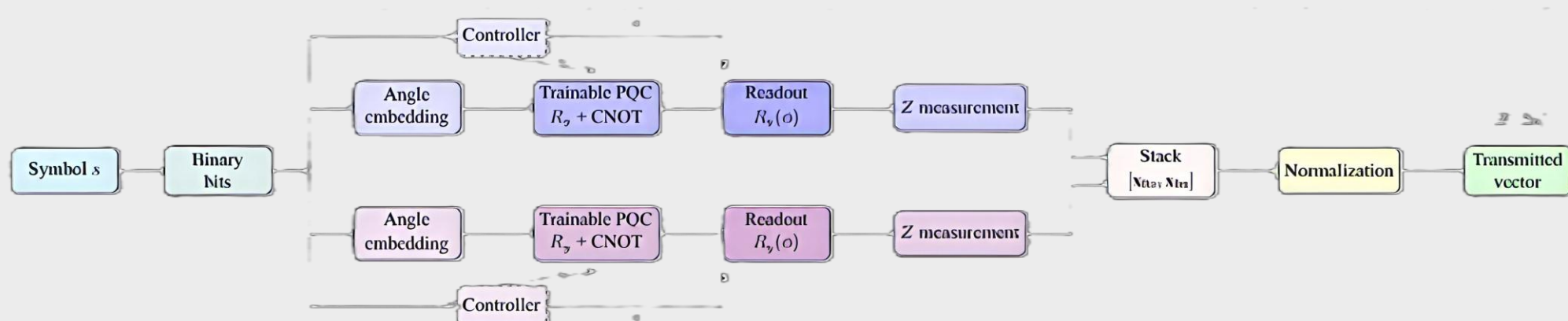
Real and imaginary PQC outputs stacked and normalized



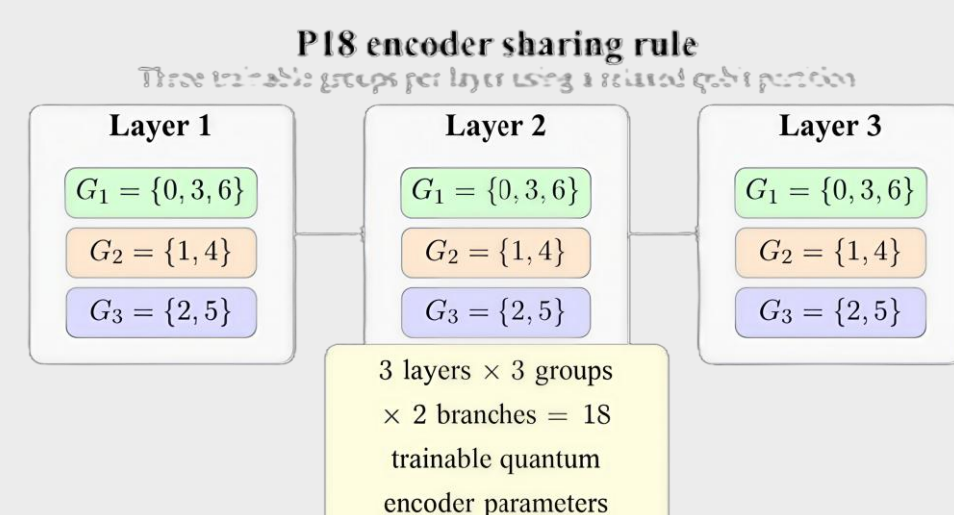
Main goal: minimize the difference between the transmitted message s and the decoded message s'

Custom Quantum Encoders:

1. Quantum Encoder with lightweight classical controller



2. P18 model



Results

1. HAE vs CAE for linear channel

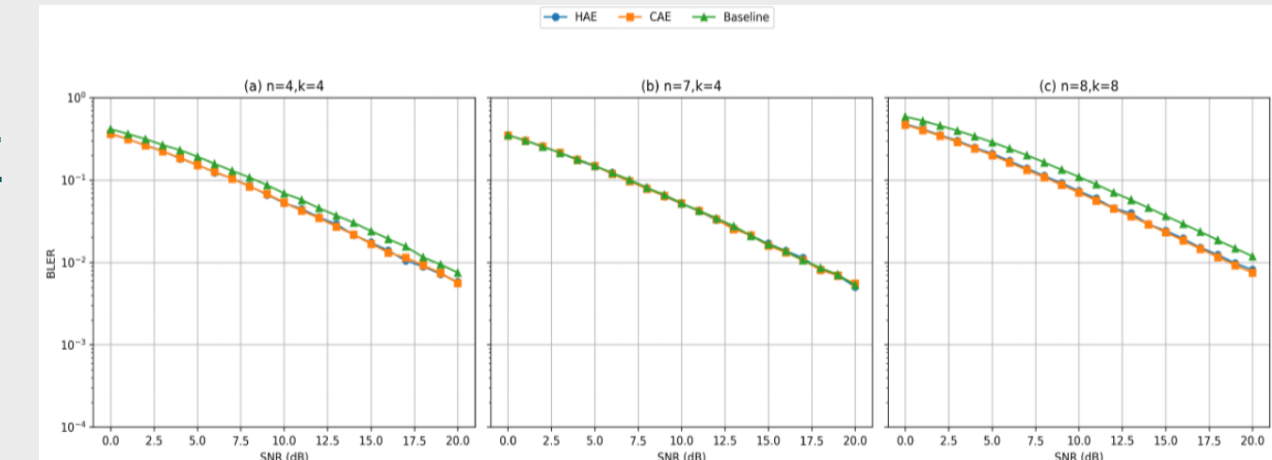
1.1 Parameter Saving

Approximately 60%–70% fewer trainable parameters.

Scheme	(4,4)	(7,4)	(8,8)
CAE	1232	1532	209952
HAE	440	554	70192

1.2 Performance Evaluation

- Similar BLER between HAE and CAE

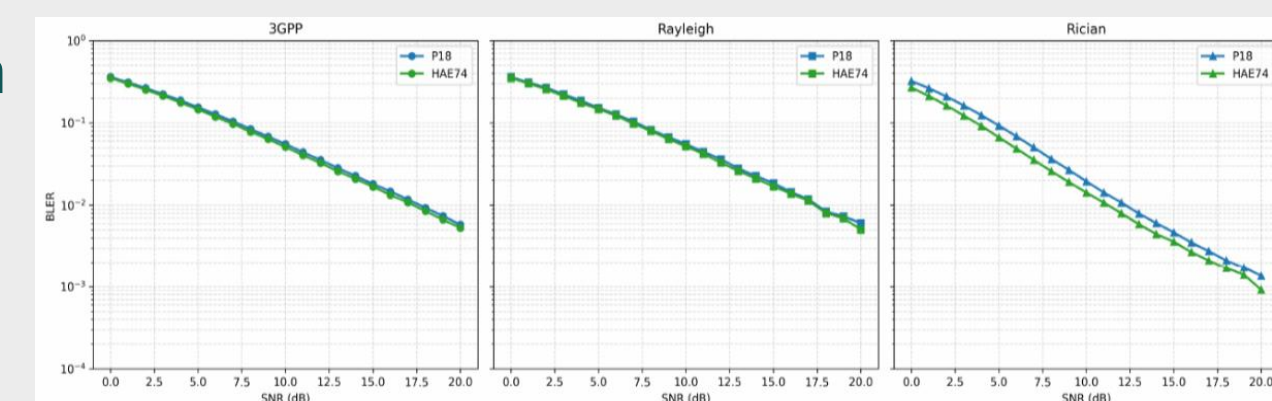


- Same trend for Rayleigh, Rician and 3GPP channels

2. P18 model vs Baseline HAE

2.1 Performance Evaluation

- Rayleigh and 3GPP: P18 same BLER trend as the original HAE.



- Rician: Gap becomes more visible, with the original HAE.

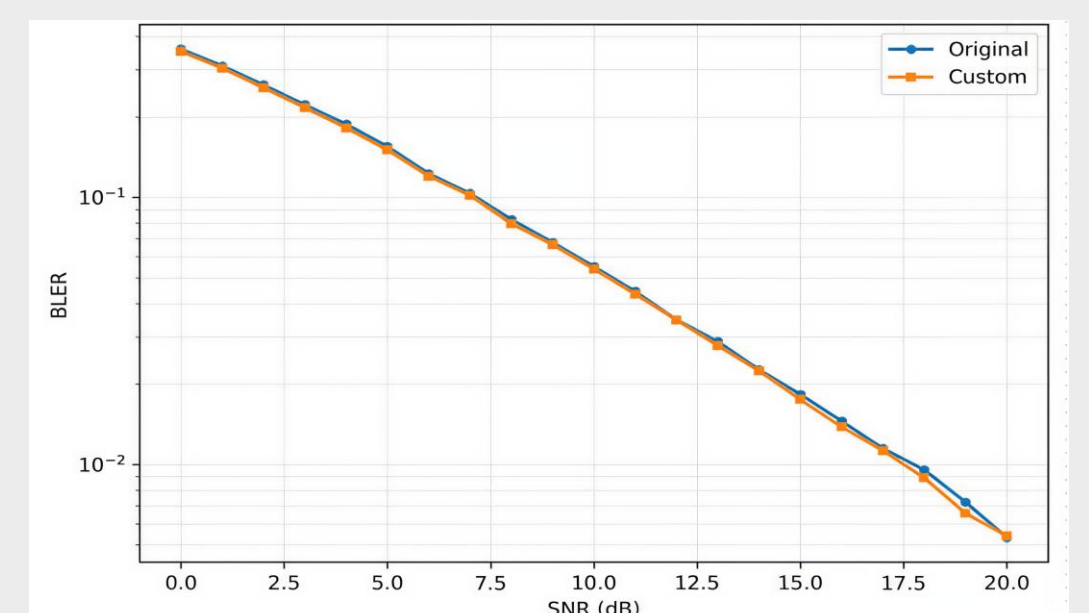
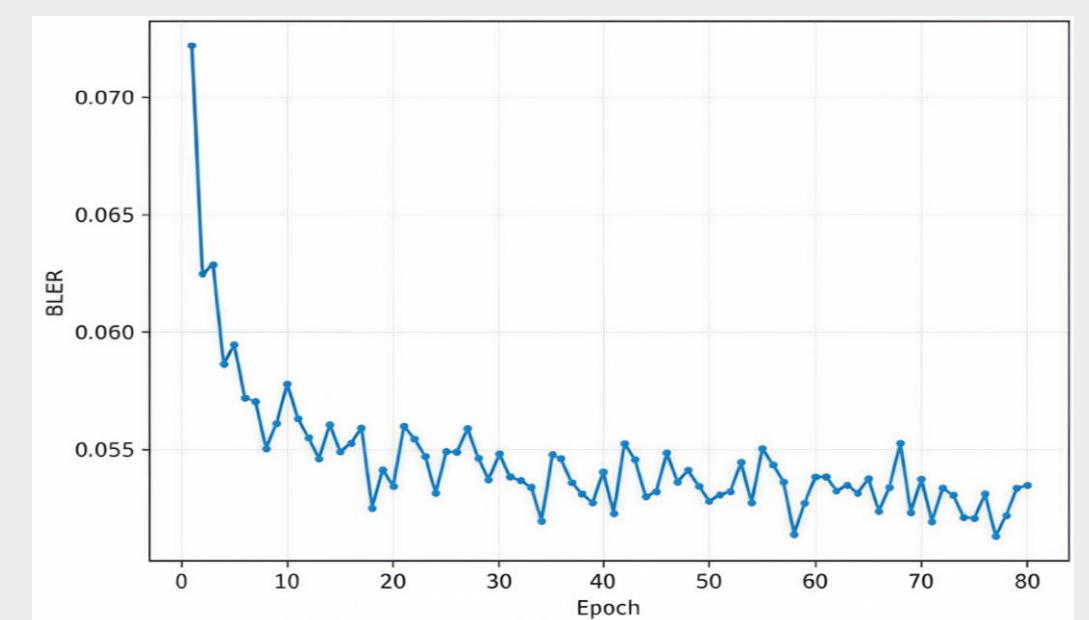
- P18: a balanced low-parameter variant.

3. Custom Quantum Encoder

3.1 Parameters Saving
22 parameters below the previous design

3.2 Performance Evaluation

- Clear learning for all training epochs
- For all SNR region: the custom encoder has slightly lower BLER.



4. Classical vs Hybrid UR-SWIPT for non-linear channel

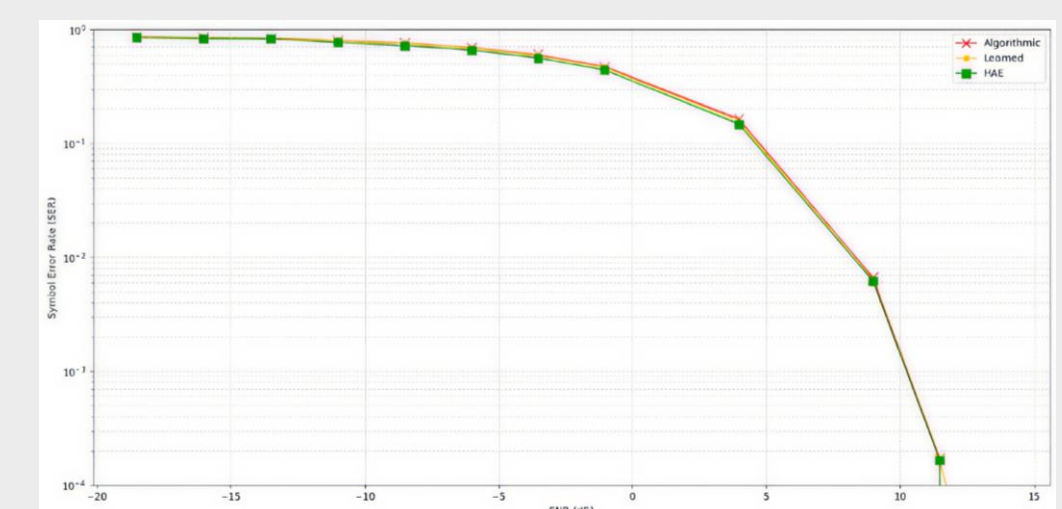
4.1 Parameters Saving

Overall parameter reduction of 37.3%.

Scheme	Encoder	Decoder	Total
CAE	81	88	169
HAE	18	88	106

4.2 Performance Evaluation

- HAE achieves slightly lower SER
- Reliable ID performance while achieving a more parameter-efficient representation.



Conclusion and Future Work

Key Findings

- Enormous parameter saving
- Similar performance between HAE and CAE schemes

Future Work

- Creation of a custom PQC layer
- Extension into the EH side of UR-SWIPT
- Further exploitation into non-linear channels

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